Open Name Tags (ONT)

A short, human-readable name — like `alice` — settled on Bitcoin, that you actually own. design brief · signet prototype

₿1 = 1 satoshi · ~$ at ~$100k/BTC

ONT is a global namespace of names settled on Bitcoin — names are acquired once: a sunk **₿1,000** (~$1) miner-fee gate for the long tail, batched thousands-per-anchor so it scales; a bonded on-chain auction when a name is actually contested. After that, one owner key controls the name, and anyone can re-derive who owns what from Bitcoin plus public batch data — servers get checked, not believed. Committing so little works because the chain fixes the only things that need global ordering — who claimed or bonded which name, and by when; everything mutable (what a name *points to*) is owner-signed off-chain data that needs a signature, not a block.

The no-list: no token · no new chain · no governance body or vote · no rent, renewal, or expiry · no reserved names, founder allocation, or admin key.

THE TRUST MODEL — WHO CAN SCREW YOU, AND HOW THAT'S STOPPED

What you must trust: Bitcoin, for ordering and final settlement (ONT adds no consensus of its own). **A small consensus core** — three files, CI-locked so the audited surface can't silently grow; honest scope: today they determine owner-key authority and replay validation, while auction winner-becomes-owner still sits outside the boundary (decided to move inside, gated on demonstrated correctness). And **pre-launch, the project itself** — the rules are designed to be frozen at launch, but they are **not frozen today**: several parameters are placeholders, flagged below.

YOU DON'T TRUST...	...BECAUSE
Publishers (write-side)	They can't decide ownership (Bitcoin does) and can't take an *existing* name (first-anchor-wins at replay). A publisher *can* misbatch a new claim — wrong key, omission — but misbehavior is public, the recourse is on-chain under the normal claim/bond rules, and the loss is bounded at ~$1.
Resolvers (read-side)	Every answer traces to an anchored root and owner signatures. A lying resolver fails re-verification — caught, not obeyed (light-client caveat below).
The founder	No admin key, no reserved names, no token, no rent stream; the gate pays miners, not the project. Anyone can run the infrastructure.

WHERE THE MONEY GOES

YOU PAY	AMOUNT	WHO ENDS UP WITH IT	COMES BACK?
Claim gate	**₿1,000** (~$1)	**Bitcoin's miners** — the anchor's fee must cover the sum of its per-name gates, so batching can't discount it	No — sunk by design (anti-spam without rent)
Publisher service fee	thin markup, TBD	the publisher	No — but capped by the standing alternative of claiming directly on L1
Auction bond	**₿50,000** floor (~$50), more if outbid	**nobody** — your UTXO throughout, posted to maturity	**Yes** — released after ~1 year; the real cost is carry, not the headline
Transfer / recovery	normal L1 fees	Bitcoin's miners	No

No protocol revenue, no token, no rent stream — nothing to extract with. The gate is fixed in bitcoin, not pegged to a dollar (a peg would need a trusted price feed); its ~$ value drifts, and we accept that drift deliberately.

THE ATTACK TOUR

- **"I'll squat everything."** The gate is **sunk, not rent** — ~10 BTC (~$1M) per million names, no rent income, no expiry to flip into renewal revenue. The valuable head doesn't go cheap: any wanted name can be bonded into an auction during its notice window (**largest bond wins**), and ≤4-char names carry a mandatory opening bond — **₿100,000,000** (1 BTC, ~$100k) at 1 char, halving per character; even there the real cost is carry (~$5,000/yr at 5%), since the principal returns. Honest residual: long-tail names nobody contests can be squatted at ₿1,000 each. Squatting isn't impossible; it's priced.
- **"I'll front-run your claim."** Ordering can't award a name — not even a miner's. Two claims with no bond **nullify**: no owner, the name reopens. A claim with no bond can deny, never award; outcomes are deadline-derived, not order-derived. There is no MEV to extract from the claim path.
- **"Then I'll grief you with collisions forever."** Each denial round burns the attacker a fresh sunk ₿1,000. The defender exits once: post a qualifying bond and the griefer must out-bid with real capital locked ~1 year — modeled, the attacker's sunk spend exceeds the defender's carry at every phase of the window; **one bond ends the game**. The disclosed asymmetry: defense takes bond-floor capital. We document that rather than paper over it — no sponsorship or proxy-bonding tooling, in v1 or as a protocol direction; third-party bonding is already permissionless, and a loan is arranged outside the protocol.
- **"The publisher steals my name — or my dollar."** It can't decide ownership by fiat — Bitcoin does — and it can't take a name you already own (first-anchor-wins at replay). It *can* commit a wrong owner key or pocket a payment on a new claim; that misbehavior is public, and your recourse is on-chain: re-claim via another publisher or L1, or contest with a bond under the normal largest-qualifying-bond rule. Worst case ≈ ₿1,000 + the thin fee, ~$1. The flow is deliberately **pay-first with reputable publishers**; with ~$1 at risk, atomic payment↔inclusion binding is later research, not a v1 dependency.
- **"The resolver lies to me."** Verify-don't-trust: the indexer re-verifies every batch membership proof against the on-chain anchored root; value records are owner-signed and sequence-linked — servable, not inventable. **Honest gaps:** producers don't emit `bitcoinInclusion` proofs yet, so a phone today trusts the resolver it queries; and auction proof bundles can't yet prove bid-set *completeness* against L1 — the same light-client work closes both.
- **"The publisher withholds the batch bytes."** The designed rule is **fail-closed**: bytes not public by a Bitcoin-height-keyed deadline don't count — withheld claims are excluded, never awarded. **Honest: the sharpest open item** — the deadline is design + simulation only; today's loop re-verifies and retries. Bytes are content-addressed against the anchored digest, so anyone can mirror them and transport is not consensus-critical.
- **"A whale sweeps the launch."** No pre-grab, no reserved list; claiming opens at a pre-announced height, every swept name is bondable into a public auction for the whole notice window, and short names carry the opening bonds above. Stated plainly: is a long window enough, or does launch need a

decaying gate? Capital-rich actors are most useful as watchtowers and contest backstops, not allocators.

- **"This is OP_RETURN spam."** Issuance amortizes to **~0.015–0.019 vB**/name at ~10k claims per anchor. The gate can't be batched away: an anchor counts only if its miner fee ≥ the sum of per-name gates. Events are single OP_RETURNs up to ~171 bytes — above the 80-byte default datacarrier policy, relying on modern node policy (confirmed relaying on signet); an explicit ask below.
- **"Why not BIP-353 / ENS / Namecoin?"** BIP-353 / Lightning Address / NIP-05 are domain-bound: lose the DNS name, lose the handle. ONT carries the same BIP-21/BIP-353-shaped payment payload and swaps only the resolution root. ENS charges rent under a token DAO; Namecoin's first-come-free invited squatting on a separate chain. ONT: no rent, no token, no governance body, priced contention, settled on Bitcoin itself.
- **"Who runs this, and what's the exit-scam?"** Today: one founder-run publisher + resolver stack on a private signet; discovery is config-seeded (registry-free on-chain scan designed, not built). Anyone can self-host both. No token to dump, no rent stream; in-flight exposure is ~$1 per pending claim. If all project infrastructure vanished, ownership re-derives from Bitcoin plus the mirrorable batch bytes — subject to the data-availability caveat above.

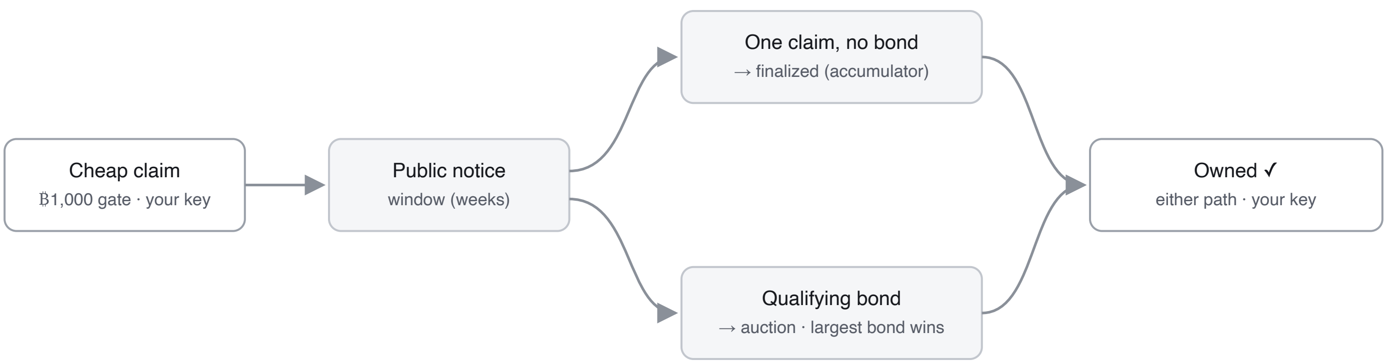
WHAT TOUCHES BITCOIN, WHAT DOESN'T

Lightning (off-chain): paying for a claim — the Ƀ1,000 gate funneled to miners + the thin publisher fee. **Bitcoin (on-chain):** one ~73-byte OP_RETURN anchor (`prevRoot → newRoot`) per ~10k claims; bonded auctions for contested names; transfers and recovery — the rare, high-stakes events. **Off-chain, free, forever:** pointing your name anywhere (owner-signed value records — instant, no fee, no chain touch). The notice window itself costs nothing: uncontested → finalized and resolvable. Try it live on signet: claim.opennametags.org — keys generated in your browser, claim batched and anchored, name visible in the explorer once the anchor confirms.

THE ROLES, DEFINED BY WHAT THEY CANNOT DO

	PUBLISHER (WRITE)	RESOLVER (READ)	WATCHTOWER (RECOVERY VETO)
Does	Accepts paid claims (Lightning), batches ~10k into one Merkle root, anchors it, serves the batch bytes (`/da/{root}`)	Replays Bitcoin + batch data through the consensus core; serves ownership, owner-signed records, proof bundles	Watches names it guards; cancels a malicious recovery within the challenge window
Cannot	Mint or move ownership; lock you out (direct L1 self-claim always works)	Decide or invent ownership; forge records	Move, transfer, or point a name — **abort-only** by construction
Worst case	You're out ~Ƀ1,000 (~$1) and re-claim elsewhere	A wasted query — verify, compare resolvers, or self-host	It goes offline and merely adds nothing
Status	Live (signet), single-writer; multi-publisher simulated	Live (signet)	Designed, not built — the credential construction is an open problem

Recovery is opt-in: a name with no recovery descriptor is one key, cold-storage style. Publisher and resolver are separate at the protocol layer (claim via A, verify via B, self-host either) even when shipped as one operator stack.



Either path ends in the same place — a name controlled by your key. Two or more claims with no bond **nullify** — no owner, the name reopens; a claim with no bond can deny, never award. **Bond-first** is allowed for known-premium names. An auction winner owns at **settlement** — the bond stays posted ~1 yr then returns, an ONT replay rule (break it early and you forfeit the name), not a Bitcoin timelock.

THE NUMBERS (PLACEHOLDERS FLAGGED — THEY SET REPLAY BEHAVIOR, SO THEY FREEZE AT LAUNCH)

PARAMETER	VALUE	STATUS
Claim gate, every name	**Ƀ1,000** ~$1, sunk, to miners	baseline
Publisher service fee	thin markup, TBD (Ƀ200 signet demo value is a placeholder, likely too high)	placeholder
Auction bond floor	**Ƀ50,000** ~$50, returnable	placeholder · load-bearing
Bond maturity	~52,560 blocks (~1 yr)	placeholder / test override
Notice window	target weeks (test: 6 blocks)	placeholder · fairness lever

Opening bond for scarce short names (≤4 char).

NAME LENGTH	OPENING BOND	~USD
1 char	**Ƀ100,000,000** (1 BTC)	~$100k
2 char	Ƀ50,000,000	~$50k
3 char	Ƀ25,000,000	~$25k
4 char	Ƀ12,500,000	~$12.5k
5+ char	gate only; Ƀ50,000 floor if contested	~$1 / ~$50

Least sure of: the real-world contest rate (high on the obvious head, low across the long tail), and whether the launch notice window is long enough for a competitive early market to form — so premium names aren't swept cheaply before other bidders show up.

Data-availability windows	unset	open
On-chain footprint	~0.015–0.019 vB/name @ ~10k/batch	measured
OP_RETURN event size	up to ~171 bytes	measured

STATUS — HONEST (SIGNET PROTOTYPE; NOT MAINNET-READY)

Live end-to-end on a private signet: owner-key transfer, owner-signed value records, recovery, a bonded auction bid the resolver accepts — and, since 2026-06-09, the full batched claim path: a claim at claim.opennametags.org is batched, anchored on-chain, re-verified by the indexer against the anchored root, and appears in the public explorer. One 12-word phrase derives identical keys across the claim site, web tools, and the mobile app, locked by shared conformance vectors. **Not live (disclosed):** the fail-closed data-availability deadline (design + simulation only); the light-client emit path; multi-publisher (simulated; production is single-writer); real Lightning payment (stubbed on signet); auction settlement inside the audited core (decided, gated on demonstrated correctness). Parameters are placeholders — the rules are designed to freeze at launch and are **not frozen today**.

WHAT WE MOST WANT PUSHED ON

- **The data-availability rule and transport** — is fail-closed-by-height sound against withholding and reorgs? The availability marker is now **folded into the anchor** (marker-fold (#47)) — does that miss a consensus role for a second timestamp? Is publisher-served + voluntary mirrors enough for v1?
- **Publisher trust-minimization** — is pay-first + reputable operators the right v1 stance, or is there a *deployable-today* way to bind payment to inclusion atomically?
- **The bond floor** — ₿50,000 is the price of escalation *and* of defense; a better point on the grief-vs-access curve, given the accepted asymmetry?
- **The anti-spam sink** — the gate goes to miners as fees; the right sink, or does anything argue for a provable burn instead?
- **OP_RETURN footprint** — are ~171-byte events acceptable on mainnet, or is a script/covenant carrier worth a soft-fork dependency?
- **Light-client verification** — launch blocker, or post-launch?
- **Auction form** — open ascending (current) vs sealed second-price, given MEV and relay-bid timing?
- **Launch fairness** — is a long notice window enough against a day-one sweep, or do we need a decaying launch gate?
- **The watchtower credential** — the cleanest name-scoped, abort-only construction for an unattended recovery veto?

Repo: github.com/deekay/ont · deeper: `docs/DESIGN.md` · exact lifecycle: `docs/spec/ONT_ACQUISITION_STATE_MACHINE.md` · plain language: `docs/ONT.md` · canonical status + numbers: `docs/core/STATUS.md`. Parity-bound to `docs/ONT.md`: if this page and ONT.md disagree, ONT.md wins.